MILITARY STANDARD

MIL-STD-1322-1000
(NAVY)
27 AUGUST 1981

UNIT LOADS
FOR DOMESTIC AND OVERSEAS SHIPMENT

TYPICAL LOADS

1. SCOPE

1.1 SCOPE. THIS SLASH NUMBERED DOCUMENT PROVIDES GUIDANCE TO ENABLE AN ACTIVITY TO CONSTRUCT APPROVED UNIT LOADS WHEN NO SPECIFIC SLASH NUMBER DOCUMENT EXISTS FOR THE ITEM(S) TO BE PALLETIZED.

1.2 RESTRICTIONS. THE USE OF THIS SLASH NUMBERED DOCUMENT IS LIMITED TO CONVENTIONAL AMMUNITION AND EXPLOSIVE ITEMS PACKAGED IN APPROVED CONTAINERS. IT SHALL NOT BE USED FOR LOOSE OR IRREGULAR SHAPED ITEMS THAT REQUIRE SPECIAL PURPOSE PALLETS OR METAL PALLET ADAPTERS TO CONSTRUCT THE UNIT LOAD.

1.2.1 THE USE OF THIS SLASH NUMBERED DOCUMENT IS LIMITED TO THE OCCASIONAL REQUIREMENT TO PALLETIZE ITEMS NOT COVERED BY SPECIFIC MIL-STD-1322 SLASH NUMBERED DOCUMENTS. RECURRING REQUIREMENTS REQUIRE THE DEVELOPMENT OF A SPECIFIC MIL-STD SLASH SHEET. REQUESTS FOR A SPECIFIC SLASH SHEET SHOULD BE SENT TO COMMANDING OFFICER, NAVAL WEAPONS STATION EARLE, COLTS NECK, N.J. 07722 ATTN: NAVAL WEAPONS HANDLING CENTER (NWHC).

2. DOCUMENT USE REQUIREMENTS

2.1 GENERAL. THE REQUIREMENTS OF THIS SLASH NUMBERED DOCUMENT AND THE GENERAL DOCUMENT MIL-STD-1322 (NAVY) SHALL BE FOLLOWED.

2.1.1 USE OF THIS SLASH SHEET. THE EXAMPLES SHOWN IN THIS SLASH NUMBERED SHEET ARE TYPICAL PROCEDURES. CONSTRUCTION OF THE UNIT LOAD FOR THE ITEM(S) TO BE PALLETIZED SHOULD FOLLOW THE GENERAL PRINCIPLES AND REQUIREMENTS OF THE TYPICAL UNIT LOADS THAT ARE SHOWN ON THE FOLLOWING PAGES. THE TYPICAL UNIT LOAD FOR AN ITEM WHICH MOST CLOSELY RESEMBLES THE ITEM TO BE PALLETIZED SHOULD BE USED.

2.1.2 TYPICAL UNIT LOADS NOT USEABLE. WHEN THE ITEMS SHOWN AS TYPICAL UNIT LOADS ARE NOT SUFFICIENTLY LIKE THE ITEM TO BE PALLETIZED, THE PUBLISHED MIL-STD-1322 (NAVY) DASH SHEETS (INCLUDING ALL WR-53 SLASH SHEETS NOT SUPERSEDED) SHOULD BE EXAMINED AND THE ONE WITH THE ITEM MOST CLOSELY RESEMBLING THE ITEM TO BE PALLETIZED SHOULD BE USED.

* U.S. GOVERNMENT PRINTING OFFICE: 1981—505-022/7398

FSC 8140

AUTHORIZED AND RELEASED

FOR THE PREPARATION OF UNIT LOADS FOR DOMESTIC AND OVERSEAS SHIPPING, HANDLING AND STORAGE.

M.L. Gray NWHC 12/12/79
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

E. Kelly 12/13/79
SIGNATURE SEA SYSCOM BY DIRECTION DATE

REQUIREMENTS FOR CONSTRUCTION OF THIS UNIT LOAD SHALL CONSIST OF THIS DOCUMENT AND THE LATEST ISSUE OF MIL-STD-1322 (NAVY)

ORIGINATOR D. McCloskey 11/27/79
DATE

NAVAL WEAPONS HANDLING CENTER
WPNSTA EARLE, NEW JERSEY

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2.1.3 ASSISTANCE. WHEN IT HAS BEEN DETERMINED THAT THE INSTRUCTIONS OF PARAGRAPHS 2.1.1 AND 2.1.2 DO NOT PROVIDE SUFFICIENT GUIDANCE, ADDITIONAL ASSISTANCE MAY BE REQUESTED FROM THE NAVAL WEAPONS STATION EARLE (NWHC), AUTOVON 449-7692, 7693, 7691.

3. MATERIAL REQUIREMENTS

3.1 GENERAL. UNLESS OTHERWISE SPECIFIED, ALL MATERIAL USED IN THE CONSTRUCTION OF THE UNIT LOADS SHALL BE AS SPECIFIED IN THE GENERAL PALLETIZING DOCUMENT MIL-STD-1322 (NAVY).

3.2 PALLETS. ONLY STANDARD SIZE WOOD PALLETS (40 x 48 AND 35 x 45 1/2 SHALL BE USED. MK 3 AND MK12 METAL PALLETS MAY BE USED WHEN OVERALL LOGISTIC COST SAVINGS OUTWEIGH THE HIGHER COST OF METAL PALLETS.

3.3 STEEL STRAPPING. UNLESS OTHERWISE SPECIFIED 3/4 INCH STRAPPING MAY BE USED FOR UNIT LOADS WITH A GROSS WEIGHT OF LESS THAN 2800 LBS. WHEN THE UNIT LOAD GROSS WEIGHT EXCEEDS 2800 LBS., 1 1/4 INCH STRAPPING SHALL BE USED.

4. CONSTRUCTION REQUIREMENTS

4.1 ITEM PLACEMENT ON PALLET. UNIT LOADS SHALL ONLY BE MADE UP IN FULL LAYERS THAT ARE CENTERED LENGTHWISE AND WIDTHWISE ON THE PALLET. THE UNIT LOAD MAY CONSIST OF ONE OR MORE LAYERS AS APPROPRIATE. QUANTITIES OF ITEMS WHICH DO NOT EQUAL A FULL LAYER SHALL BE SHIPPED, HANDLED AND STORED/STOWED LOOSE (NOT UNITIZED).

4.1.1 ITEMS SHOULD BE ARRANGED ON THE PALLET TO FORM SOLID, COMPACT LAYERS. ITEMS IN PLACE ON A PALLET SHOULD REMAIN IN PLACE WITHOUT ADDITIONAL SUPPORT. OVERHANG IN EXCESS OF 3 INCHES SHOULD BE AVOIDED.

4.2 SIZE AND WEIGHT. UNIT LOADS SHOULD NOT EXCEED 45 INCHES IN WIDTH, 54 INCHES IN LENGTH AND 47 INCHES IN HEIGHT. DIMENSIONS MAY BE INCREASED IF MANDATED BY THE SIZE OF THE ITEM AND THE LOGISTIC/SHIPPING CYCLE IS SUCH THAT NO PROBLEMS WILL BE ENCOUNTERED DURING HANDLING, SHIPPING, AND STORAGE/STOWAGE. THE GROSS WEIGHT OF THE UNIT LOAD SHOULD NOT EXCEED 3000 POUNDS.

4.3 UNDERHANG. UNDERHANG SHALL NOT BE PERMITTED. SPACERS AND BATTENS SHALL BE USED TO FILL OUT THE UNIT LOAD SO THAT THE OVERALL NOMINAL PLAN VIEW DIMENSIONS ARE FLUSH WITH THE PALLET DIMENSIONS.

4.4 TIE-DOWN STRAPS. UNLESS OTHERWISE SPECIFIED ALL UNIT LOADS SHALL HAVE A MINIMUM OF FOUR TIE-DOWN STRAPS WITH TWO STRAPS RUNNING THE LENGTH OF THE UNIT LOAD AND TWO STRAPS RUNNING THE WIDTH OF THE UNIT LOAD.

4.4.1 ADDITIONAL STRAPS SHALL BE USED AS NECESSARY TO PROVIDE POSITIVE RESTRAINT OF ALL ITEMS. THIS MAY BE ACCOMPLISHED BY USE OF ADDITIONAL TIE-DOWN STRAPS, HORIZONTAL STRAPS OR BOTH.

4.5 WOOD PALLET ADAPTERS. CONTAINERS OTHER THAN RECTANGULAR SHAPE SUCH AS CYLINDRICAL SHAPED CONTAINERS (DRUMS) OFTEN REQUIRE PALLETIZING. THIS REQUIRES THE USE OF ADAPTERS TO ASSIST IN SECURING THE DRUMS TO THE PALLET.

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4.5.1 WHEN REQUIRED, WOOD ADAPTERS MAY BE CONSTRUCTED AND USED PROVIDED THE ITEM IS NOT RESTRICTED AS SPECIFIED IN PARAGRAPH 1.2.

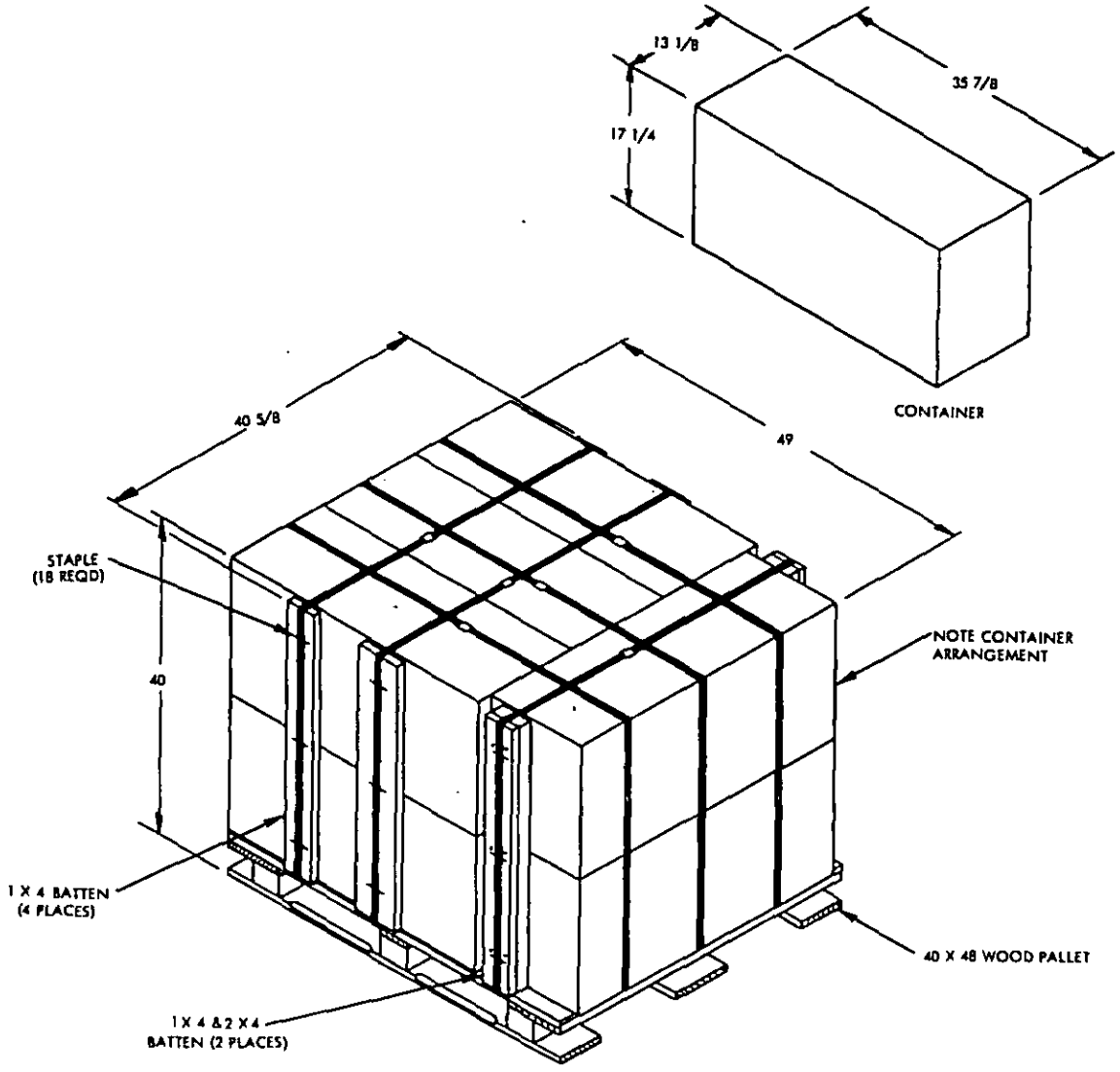
4.5.2 THE EXAMPLES USED IN THIS DASH SHEET AND OTHER MIL-STD-1322 DASH SHEETS (AND WR-53 SLASH SHEET) SHOULD BE USED TO ASSIST IN CONSTRUCTING WOOD ADAPTERS.

5. MARKINGS

5.1 THE UNIT LOAD SHALL BE MARKED IN ACCORDANCE WITH THE REQUIREMENTS OF THE GENERAL DOCUMENT MIL-STD-1322 (NAVY).

6. INSPECTION

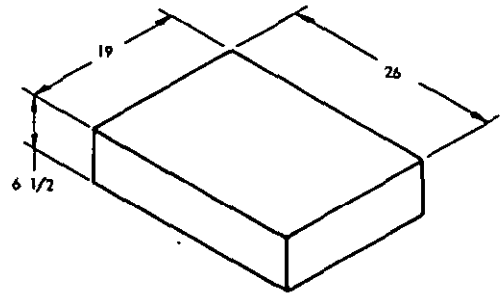
6.1 THE COMPLETED UNIT LOAD SHALL BE INSPECTED IN ACCORDANCE WITH PARAGRAPH 4.9 OF THE GENERAL DOCUMENT MIL-STD-1322 (NAVY).



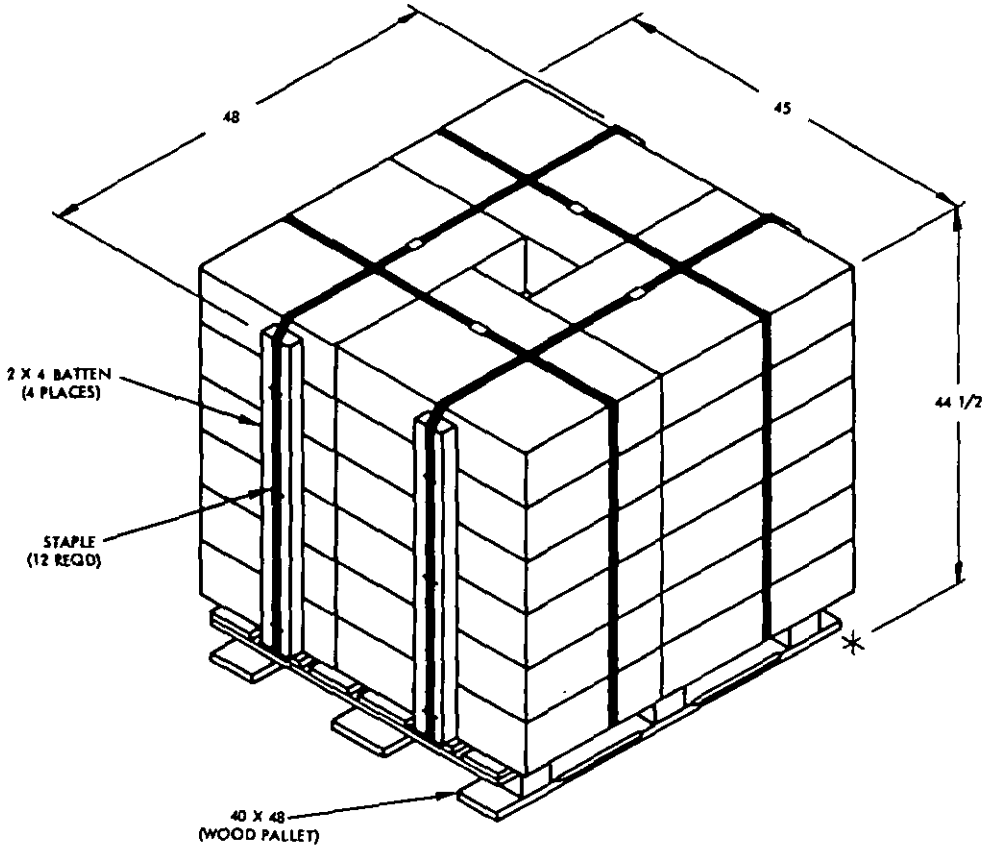
DETAIL A

UNIT LOAD WITH UNDERHANG
CORRECTED USING BATTENS (2 SIDES)
AND CONTAINERS ARRANGED AT RIGHT ANGLES

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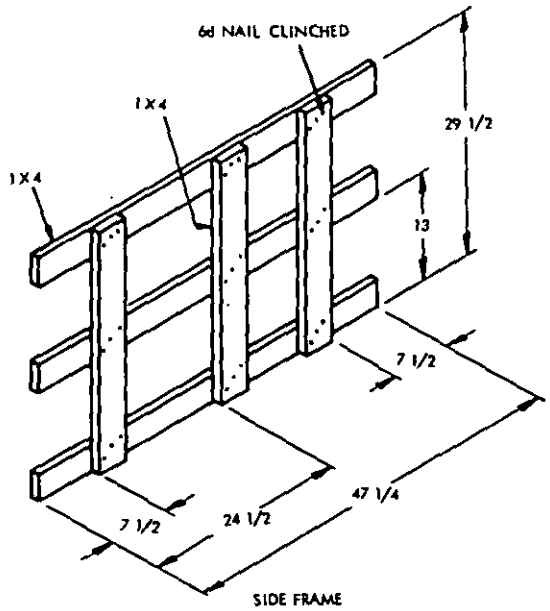
CONTAINER



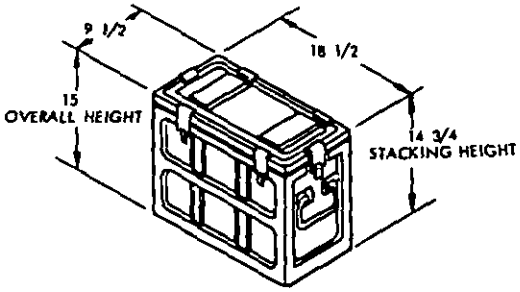
DETAIL B

UNIT LOAD WITH UNDERHANG (2 SIDES)
CORRECTED USING BATTENS
CONTAINERS ARRANGED "CHIMNEY PATTERN"

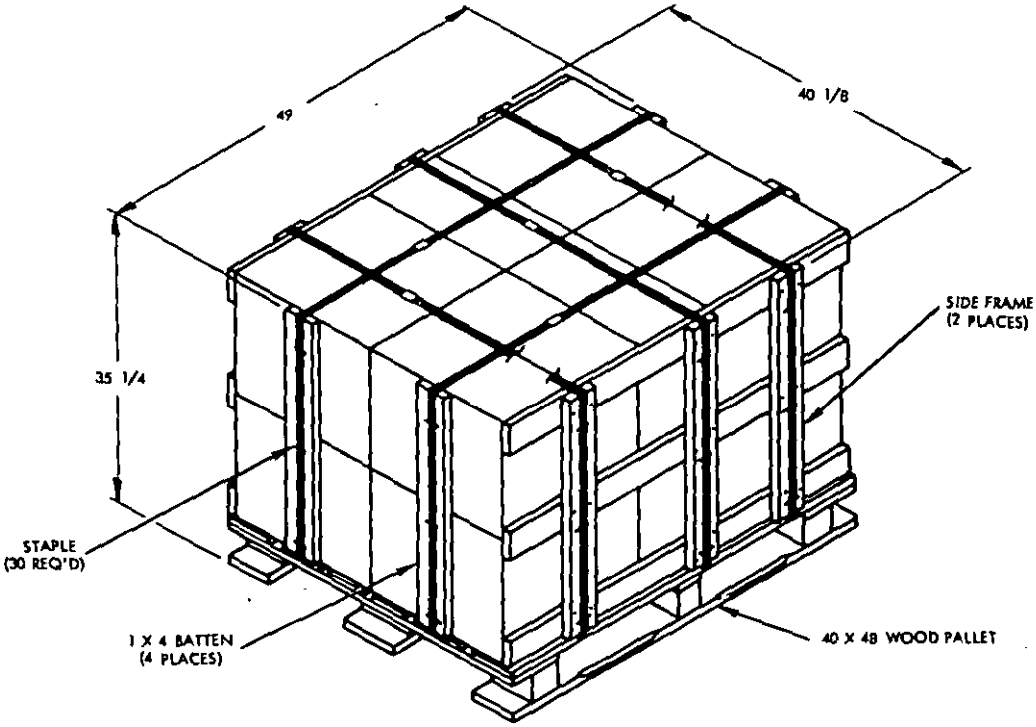
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SIDE FRAME
(2 REQ'D)



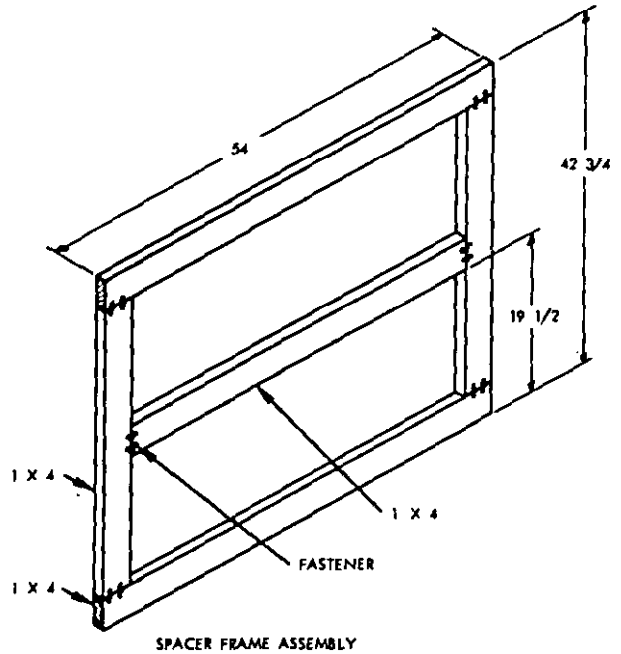
MK 1 AMMUNITION
BOX



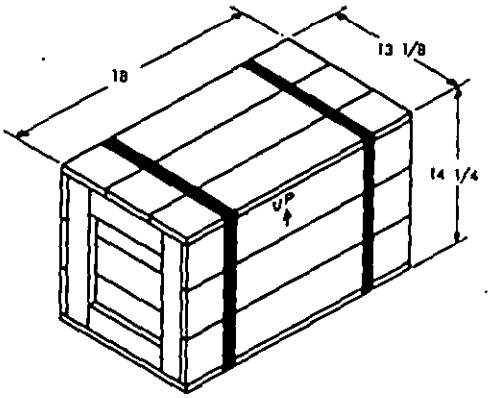
DETAIL C

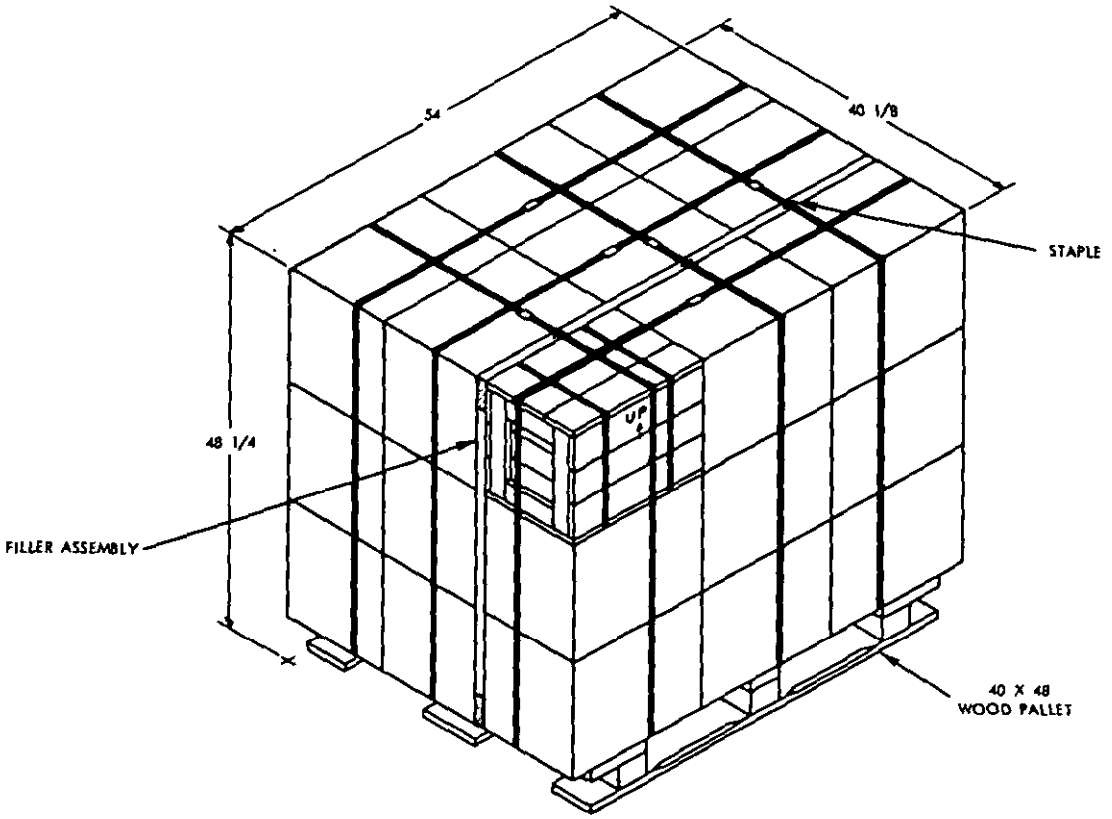
UNIT LOAD WITH UNDERHANG (4 SIDES)
CORRECTED USING BATTENS
AND SIDE FRAMES

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SPACER FRAME ASSEMBLY
1 REQUIRED

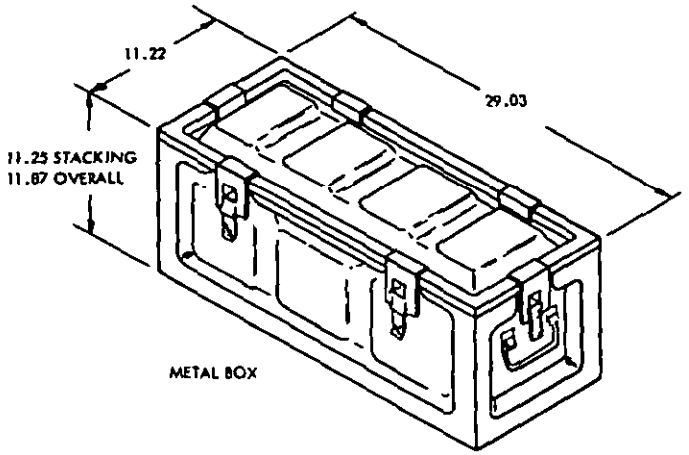


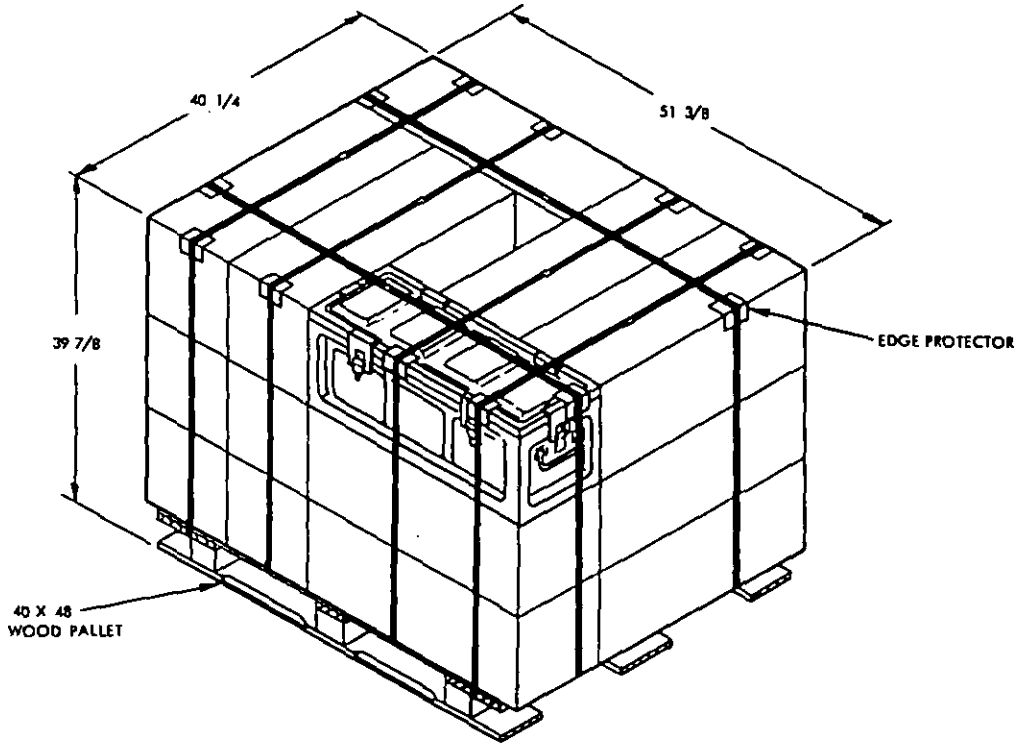


DETAIL D

UNIT LOAD WITH SPACER FRAME ASSEMBLY
USED TO PREVENT UNDERHANG

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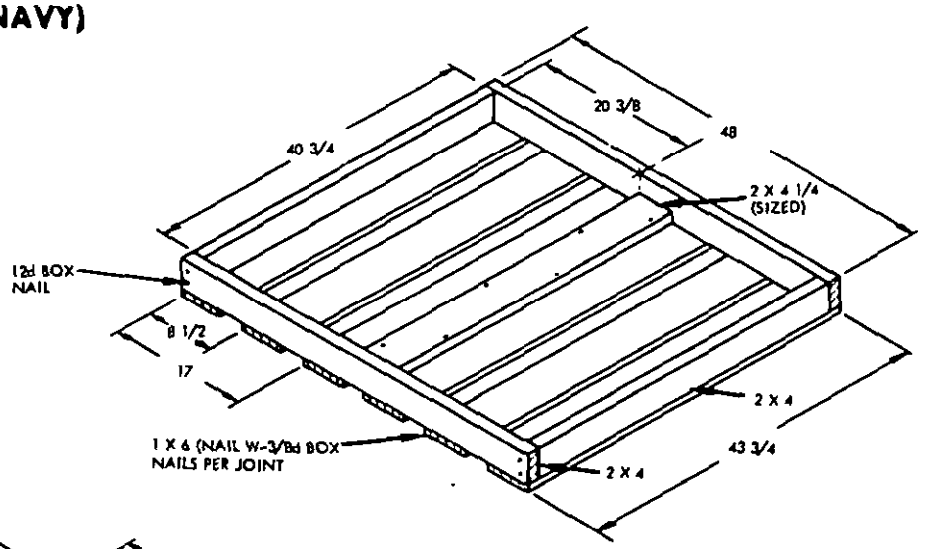




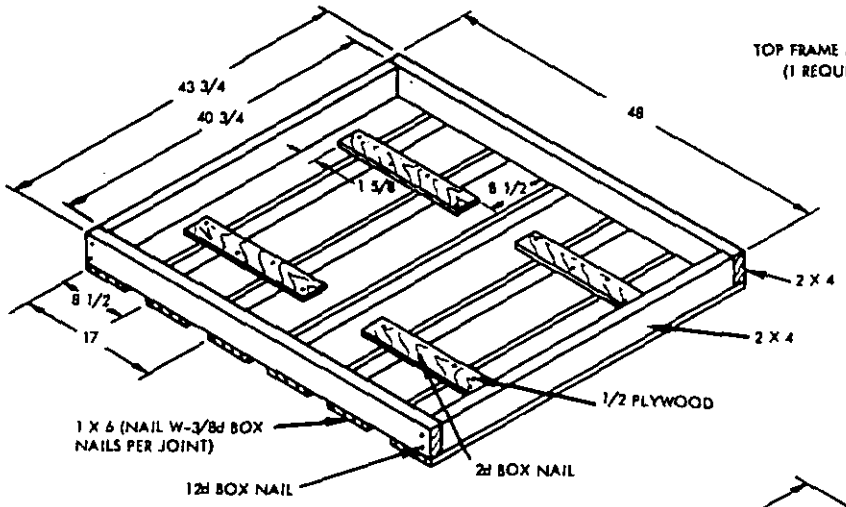
DETAIL E

UNIT LOAD WITH BOXES ARRANGED
MODIFIED "CHIMNEY PATTERN"

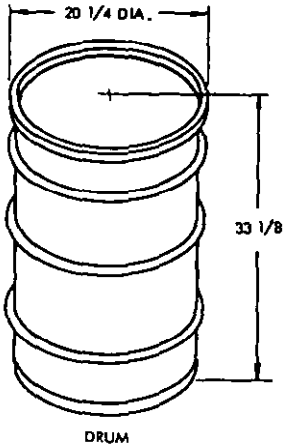
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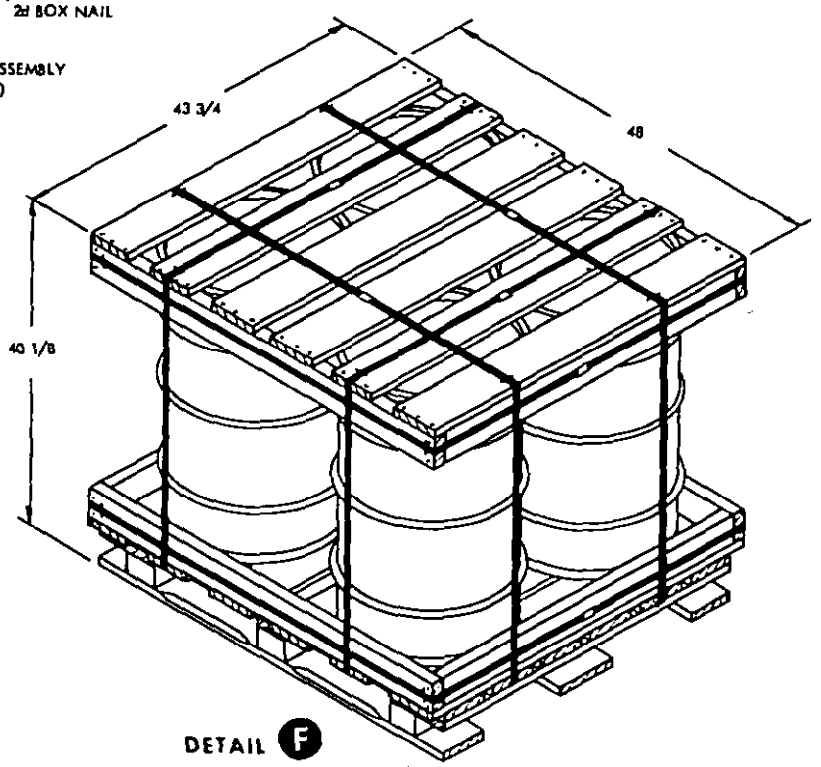
TOP FRAME ASSEMBLY
(1 REQUIRED)



BOTTOM FRAME ASSEMBLY
(1 REQUIRED)



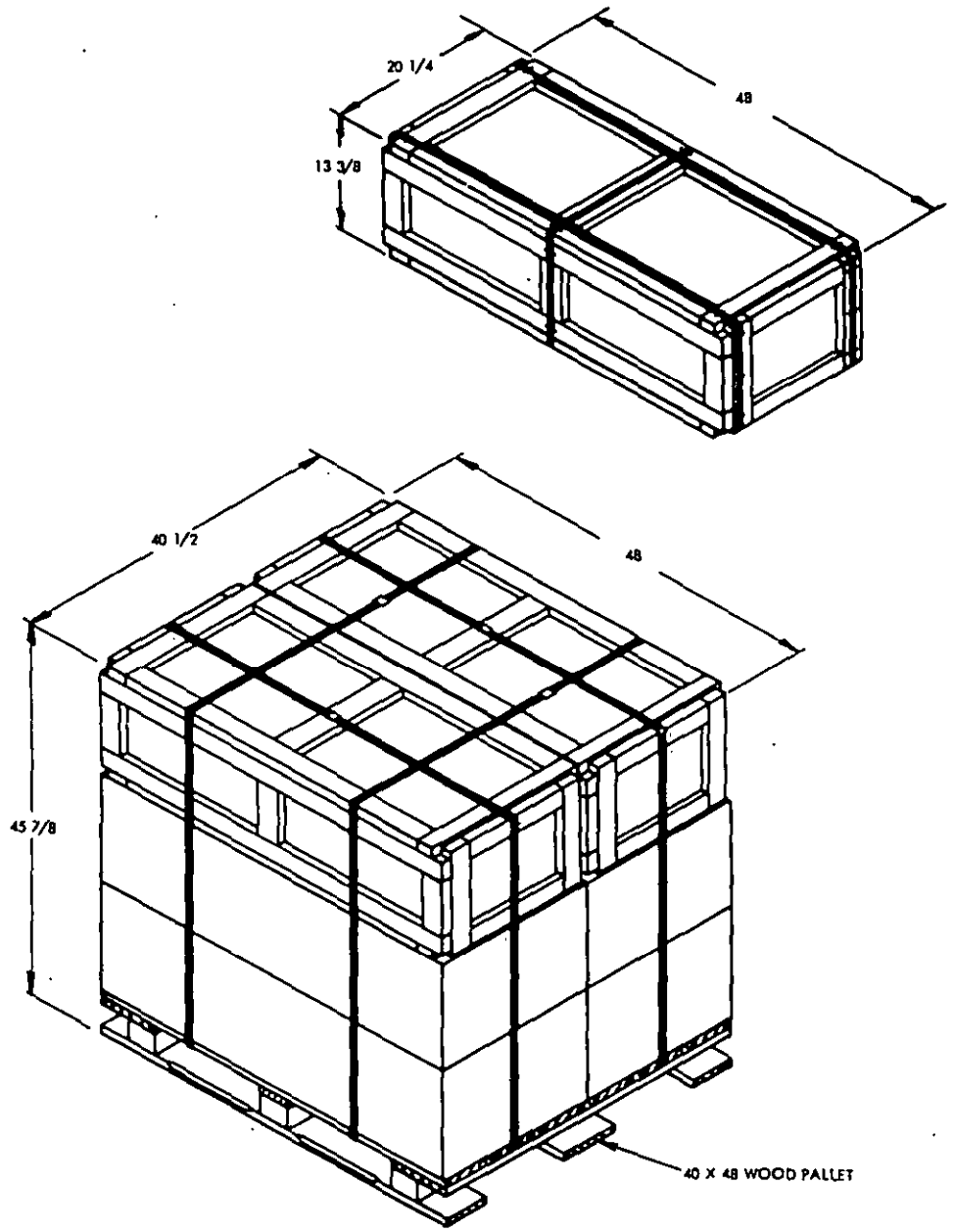
DRUM



DETAIL F

UNIT LOAD OF DRUMS
USING WOOD ADAPTERS

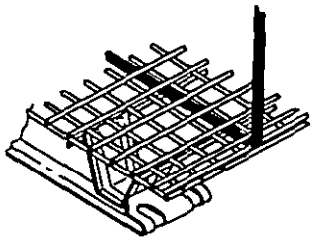
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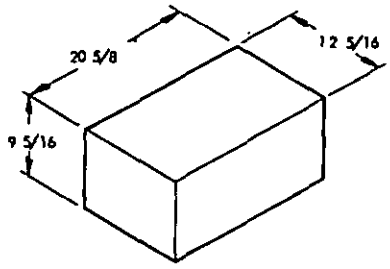
DETAIL G

UNIT LOAD WITH NO UNDERHANG

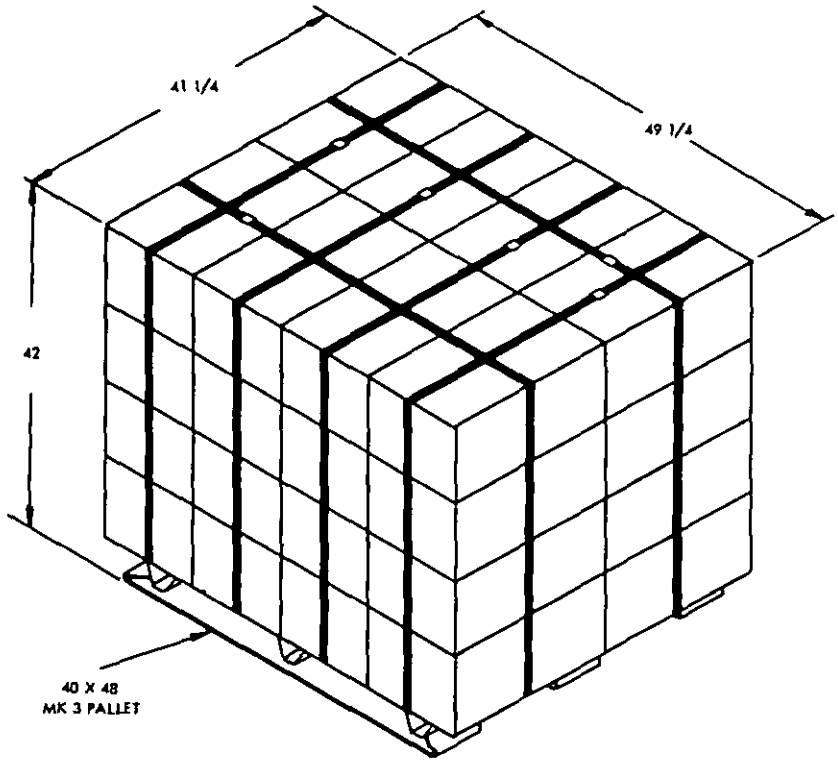
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STRAP THREADING
DETAIL



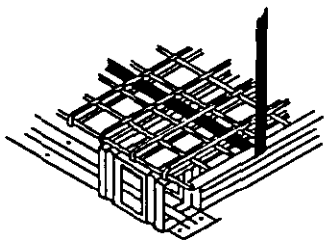
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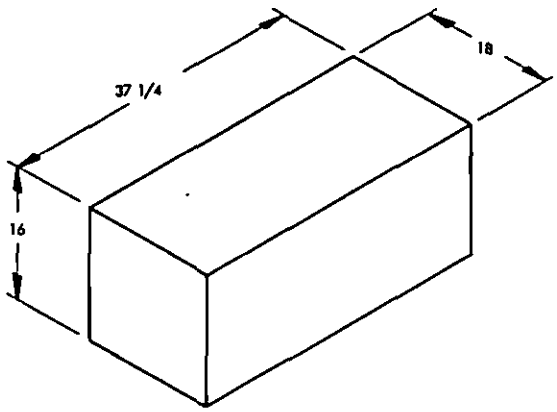
DETAIL H

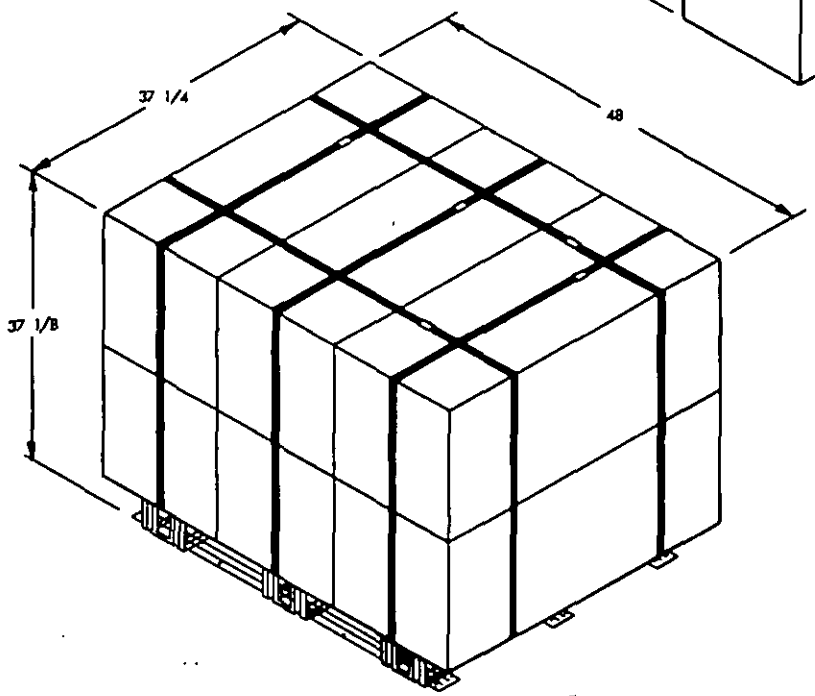
UNIT LOAD ON MK 3 METAL PALLET
(SEE PARAGRAPH 3.2)

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STRAP THREADING
DETAIL





DETAIL J

UNIT LOAD ON MK 12 METAL PALLET
(SEE PARAGRAPH 3.2)

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-STD-1322-1000 (NAVY)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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NOTICE OF VALIDATION

INCH-POUND
MIL-STD-1322/1000(OS)
NOTICE 1
14 March 1989

MILITARY SPECIFICATION

UNIT LOAD FOR DOMESTIC & OVERSEAS SHIPMENT TYPICAL LOADS

MIL-STD-1322/1000(OS) dated 27 AUG.1981, has been reviewed and determined to be valid for use in acquisition.

Preparing Activity:
Navy - OS

AMSC N/A

FSC 8140

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